Detecting fake news: binary classification of fake vs. true news based on textual characteristics of news articles

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Note: The model fitting code runs up to 5 minutes.

1 Summary

News is everywhere. In this digital era, there exists both true and fake news on the Internet, making it harder for readers and news agents to differentiate them. So, fake news detection is a technology with rising demands that helps defend information integrity. To solve this problem, applied machine learning is used for binary classification. The fitted Naive Bayes model with `MultinomialNB` as estimator and `title`, `text` and `subject` as features yields a test classification accuracy of 0.954, where the false positive and false negative rates are both low. The AP score and AUC value of the final model end up being 0.97 and 0.98 respectively, suggesting decent and satisfactory performance. Compared to the baseline dummy classifier, this Naive Bayes model classifies both true and fake news significantly better, which is reflected by consistently high and similar training, cross-validation and test scores. This machine learning model gives us confidence in its decent performance, which can potentially be deployed and used in the reality. But limitations do exist, since the collected data do not cover all possible types of news. So, when deployed in the wild, the model might perform badly in news of other realms, such as entertainment news. If more comprehensive data can be collected, then this model might handle more cases accurately.

2 Introduction

"Fake news", also known as disinformation, is deliberately misleading or false information presented as legitimate news (Service Canada 2024). Its rapid spread in recent years has becoming a pressing concern, particularly with the rise of social media platforms that accelerate the dissemination of content. These platforms are designed with the primary objective of capturing and retaining user attention for as long as possible. To that end, fake news often exploits this design to gain traction.

There is high public awareness of this issue in Canada. In 2023, Statistics Canada reported that 59% of Canadians admitted that they were "very or extremely concerned about online misinformation" in the Survey Series on People and their Communities (Bilodeau and Khalid 2024). In the United States, the concept of "fake news" grew in popularity during the 2016 presidential election. Allcott and Gentzkow (2017) estimated that false news stories were shared on Facebook at least 38 million times in the 3 months leading up to the election. Such widespread circulation of this false information can have serious consequences, including influencing public opinion and undermining trust in legitimate news sources. Thus, methods for detecting fake news may be beneficial in mitigating its spread.

This report investigates whether a machine learning model can accurately classify news articles as "fake" or "true" based on their textual characteristics, such as the article's title, subject and body content. Automated detection of fake news has the potential to mitigate the spread and impact of information, particularly in social media's fast-paced environment where information is shared instantly and at scale, from numerous sources.

3 Data

The dataset used in this analysis is the "Fake and Real News Dataset" available on Kaggle (Bisaillon 2020). It contains approximately 40,000 news articles published in the United States between 2015 and 2018. Each row in the dataset represents a news article, with a column for its title, subject, date, and body text.

The raw data from Kaggle was initially provided as two seperate datasets: one for the fake articles and one for the true articles. To prepare the data for modeling, we conducted some validation checks to make sure the data was in the expected format, with no missing observations or duplicate entries. The two datasets were then merged into one large one, with an additional column for the `target` value, indicating whether the article was labeled as "true" or "fake". To support model training and evaluation, we shuffled the rows and split the data into training (80%) and testing (20%) sets. Further validation was conducted on the training set to identify outliers and anomoulous correlations. Finally, we conducted exploratory data analysis (EDA) on the training data set to examine the distribution of the target variable and gain initial insights into the dataset.

	title	text
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Do

	title	text
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...

3.1 Data Validation

Part 1: Perform the following data validation checks: - Correct column names - No empty observations - Missingness not beyond expected threshold - Correct data types in each column - No duplicate observations

	title	text
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a cons
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender peopl
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counse
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign a
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - Preside
...	...	...
21412	'Fully committed' NATO backs new U.S. approach...	BRUSSELS (Reuters) - NATO allies on Tuesda
21413	LexisNexis withdrew two products from Chinese ...	LONDON (Reuters) - LexisNexis, a provider of
21414	Minsk cultural hub becomes haven from authorities	MINSK (Reuters) - In the shadow of disused So
21415	Vatican upbeat on possibility of Pope Francis ...	MOSCOW (Reuters) - Vatican Secretary of Sta
21416	Indonesia to buy $1.14 billion worth of Russia...	JAKARTA (Reuters) - Indonesia will buy 11 S

	title	text
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americ
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Dev
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milw
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announce
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day
...	...	...
22698	The White House and The Theatrics of 'Gun Cont...	21st Century Wire says All the world s a sta
22699	Activists or Terrorists? How Media Controls an...	Randy Johnson 21st Century WireThe majo
22700	BOILER ROOM – No Surrender, No Retreat, Heads ...	Tune in to the Alternate Current Radio Net
22701	Federal Showdown Looms in Oregon After BLM Abu...	21st Century Wire says A new front has just
22702	A Troubled King: Chicago's Rahm Emanuel Desper...	21st Century Wire says It s not that far awa

Part 2: Perform the following data validation checks on `true_df` and `fake_df`: - No outlier or anomalous values - Correct category levels (i.e., no string mismatches or single values)

	title	text
1442	Trump's Leaks To Russia May Well Have Put Ind...	Trump s ignorance, incompetence, and ego are p
10725	Republican Senator Kirk says he wants Senate t...	WASHINGTON (Reuters) - Republican Senator
2780	Massive copper mine tests Trump's push to slas...	SUPERIOR, Arizona (Reuters) - Rio Tinto's pr
6773	Patton Oswalt To Bernie Fans: 'You're A F*cki...	Comedian Patton Oswalt has a knack for calling
13638	U.S. general sees no change in Pakistan behavi...	WASHINGTON (Reuters) - The top U.S. genera
...	...	...
7807	Clinton heavily favored to win Electoral Colle...	NEW YORK (Reuters) - After a brutal week fo
15556	Polish reforms of judiciary pose threat to rul...	BRUSSELS (Reuters) - Polish reforms of its jud
17920	Factbox: Japan main parties' key election pled...	TOKYO (Reuters) - Campaigning began on Tue

	title	text
6819	Jared Fogle’s Pedophilic Texts Finally Releas...	Former Subway spokesman and outed pedophile
15905	Iran's Rouhani: Tehran-Moscow cooperation need...	ANKARA (Reuters) - Iran s President Hassan I

Part 3: Perform the following data validation checks on `complete_df`: - Target/response variable follows expected distribution - No anomalous correlations between target/response variable and features/explanatory variables - No anomalous correlations between features/explanatory variables

	title	text
1442	Trump’s Leaks To Russia May Well Have Put Ind...	Trump s ignorance, incompetence, and ego are p
10725	Republican Senator Kirk says he wants Senate t...	WASHINGTON (Reuters) - Republican Senator
2780	Massive copper mine tests Trump's push to slas...	SUPERIOR, Arizona (Reuters) - Rio Tinto’s pr
6773	Patton Oswalt To Bernie Fans: ‘You’re A F*cki...	Comedian Patton Oswalt has a knack for calling
13638	U.S. general sees no change in Pakistan behavi...	WASHINGTON (Reuters) - The top U.S. genera
...	...	...
7807	Clinton heavily favored to win Electoral Colle...	NEW YORK (Reuters) - After a brutal week fo
15556	Polish reforms of judiciary pose threat to rul...	BRUSSELS (Reuters) - Polish reforms of its jud
17920	Factbox: Japan main parties' key election pled...	TOKYO (Reuters) - Campaigning began on Tu
6819	Jared Fogle’s Pedophilic Texts Finally Releas...	Former Subway spokesman and outed pedophile
15905	Iran's Rouhani: Tehran-Moscow cooperation need...	ANKARA (Reuters) - Iran s President Hassan I

3.2 Data Preprocessing

4 Methods

4.1 EDA

```
<class 'pandas.core.frame.DataFrame'>
Index: 30903 entries, 1442 to 15905
Data columns (total 5 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   title    30903 non-null  object
 1   text     30903 non-null  object
 2   subject  30903 non-null  object
 3   date     30903 non-null  object
 4   target   30903 non-null  object
dtypes: object(5)
```

```
memory usage: 1.4+ MB
```

As we can see from this summary, the training data contains 30903 observations. Each observation represents an article with a title, text, subject, date and target label. Each of these are non-null objects.

4.2 Count of Fake vs. Real News Articles

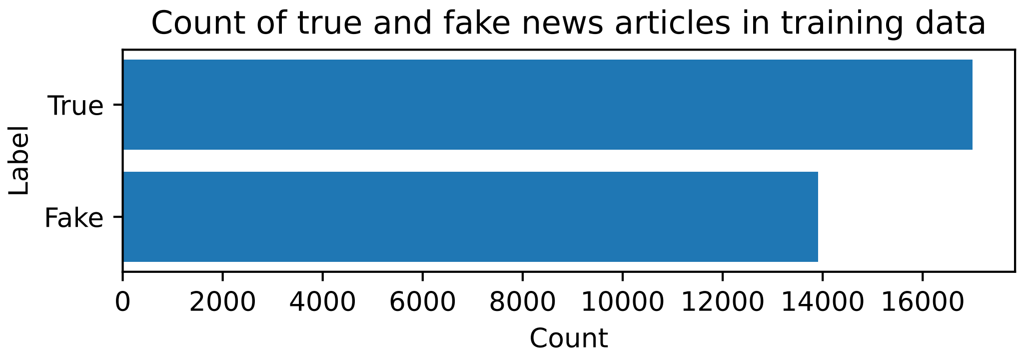

Figure 1: Count of true vs. fake news articles in training data set

From Figure 1, we can see that the training dataset is fairly balanced, with np.int64(13905) fake news articles and np.int64(16998) real news articles. This balance may help to prevent bias towards one class over the other when training our model.

4.3 Word Clouds for Titles Column

4.4 Model Building

steps	[('preprocessor', ...), ('classifier', ...)]
transform_input	None
memory	None
verbose	False

transformers	[('title_text', ...), ('text_text', ...), ...]
remainder	'drop'
sparse_threshold	0.3
n_jobs	None

transformer_weights	None
verbose	False
verbose_feature_names_out	True
force_int_remainder_cols	'deprecated'

input	'content'
encoding	'utf-8'
decode_error	'strict'
strip_accents	None
lowercase	True
preprocessor	None
tokenizer	None
stop_words	'english'
token_pattern	'(?u)\\b\\w\\w+\\b'
ngram_range	(1, ...)
analyzer	'word'
max_df	1.0
min_df	1
max_features	5000
vocabulary	None
binary	False
dtype	<class 'numpy.int64'>

input	'content'
encoding	'utf-8'
decode_error	'strict'
strip_accents	None
lowercase	True
preprocessor	None
tokenizer	None
stop_words	'english'
token_pattern	'(?u)\\b\\w\\w+\\b'
ngram_range	(1, ...)
analyzer	'word'
max_df	1.0
min_df	1
max_features	10000
vocabulary	None
binary	False

dtype	<class 'numpy.int64'>

categories	'auto'
drop	'first'
sparse_output	True
dtype	<class 'numpy.float64'>
handle_unknown	'error'
min_frequency	None
max_categories	None
feature_name_combiner	'concat'

alpha	1.0
force_alpha	True
fit_prior	True
class_prior	None

4.5 Model Evaluation

```
Classification Report:
              precision    recall  f1-score   support

        Fake       0.94      0.95      0.95      3613
        True       0.96      0.95      0.95      4212

    accuracy                           0.95      7825
   macro avg       0.95      0.95      0.95      7825
weighted avg       0.95      0.95      0.95      7825

Test Accuracy: 0.951
```

4.6 Baseline Model Comparison

```
Baseline (Dummy) Accuracy: 0.538
Naive Bayes Accuracy: 0.951
Improvement: 0.413
```

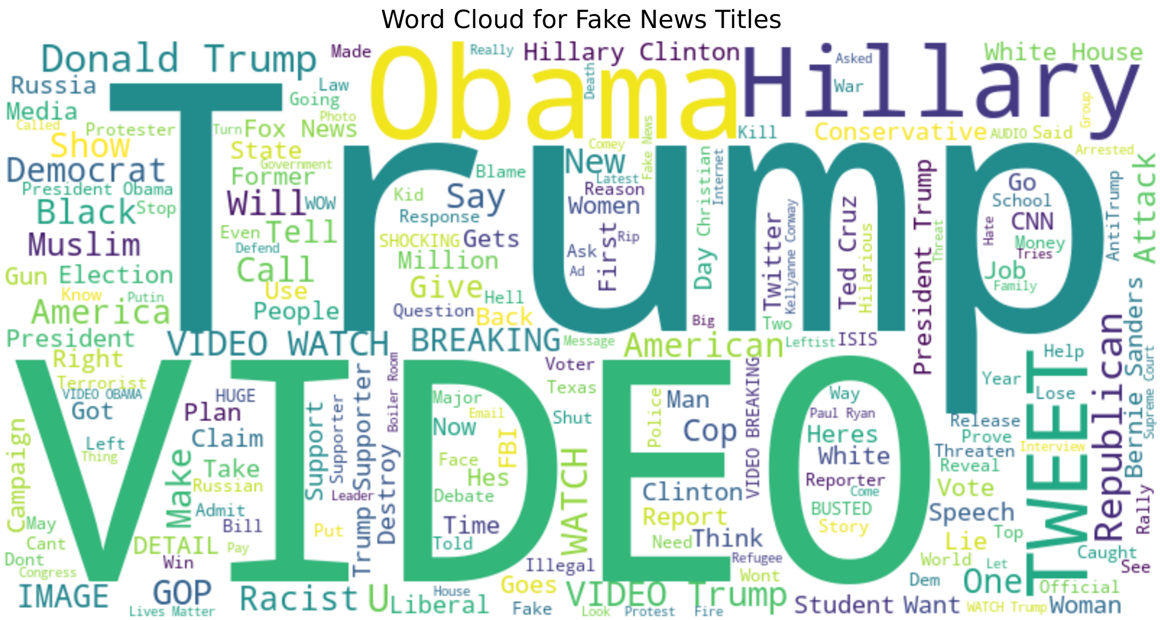

Figure 2: Word cloud for titles of fake news articles

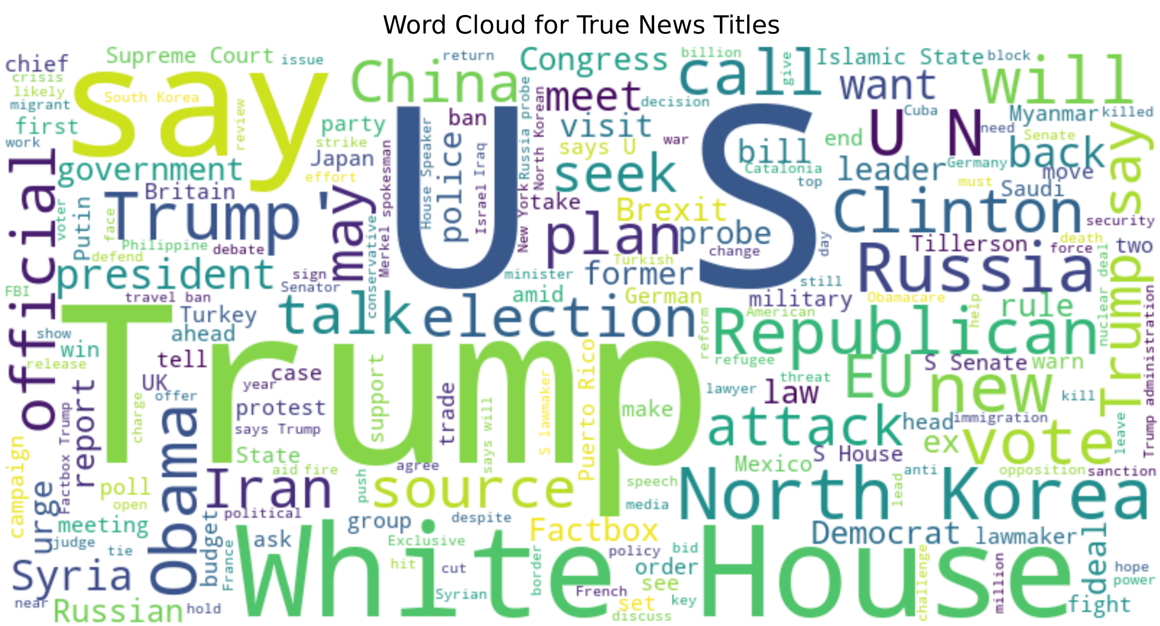

Figure 3: Word cloud for titles of true news articles

5 Results

The Naive Bayes model achieved a test accuracy of 95.1%, which represents a significant improvement over the baseline dummy classifier. The model demonstrates strong performance in distinguishing between fake and real news articles based on their textual characteristics, as shown in the word clouds (Figure 2 and Figure 3) which reveal distinct vocabulary patterns between fake and real news.

6 Conclusion

The machine learning approach using Naive Bayes classification shows promising results for fake news detection. The model's high accuracy and balanced performance across both classes suggest it could be a valuable tool in combating misinformation. However, the model's performance may be limited to the types of news articles present in the training data, and further validation on diverse news sources would be beneficial for real-world deployment.

References

Allcott, Hunt, and Matthew Gentzkow. 2017. "Social Media and Fake News in the 2016 Election." *Journal of Economic Perspectives* 31 (2): 211–36. https://doi.org/10.1257/jep.31.2.211.

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